

Paper Practice 1.6 • WORKED KEY**Descriptive Modeling and Accuracy**

Show all of your work in the space provided. When you use a metric, name it. Explanations should be one clear sentence. Circle your final answers.

1. The school store's hoodie sales for five months were 42, 55, 38, 61, and 49. (Descriptive Modeling)

a. Find the mean number of hoodies sold per month.

Answer: $245 \div 5 = 49$ hoodies

b. Name one thing the mean HIDES about these sales.

Answer: The spread – sales swing from 38 to 61, so a typical month is not reliably 49.

c. Would a table or a bar graph be more useful for spotting a trend? Explain.

Answer: A bar graph – it shows the month-to-month rise and fall at a glance, while a table lists values without showing shape.

2. Two players' season statistics are shown. (Compare Metrics)

Player	Points	Games
Alvarez	1,848	22
Brooks	1,530	17

a. Who leads in total points?

Answer: Alvarez, 1848 to 1530

b. Who leads in points per game?

Answer: Brooks: $1530 \div 17 = 90$ vs. Alvarez $1848 \div 22 = 84$

c. Which metric would you use to argue for each player? What does this show?

Answer: Total points favors Alvarez; points per game favors Brooks. Two honest metrics can rank the same data differently, so you must always say WHICH metric you used.

3. A car travels 260 miles on 8 gallons of gas. Find the miles per gallon, and explain what that metric tells a buyer. (Metrics)

Answer: $260 \div 8 = 32.5$ miles per gallon – it lets a buyer compare fuel efficiency across cars regardless of tank size.

4. A digital scale reads 9.985 grams for a mass whose true value is 10 grams. Find the accuracy of the reading. (Accuracy)

Answer: $|9.985 - 10| = 0.015$ gram

5. A bag of flour labeled 5 pounds weighs 4.92 pounds. Find the accuracy, then express it as a percent of the labeled weight. (Accuracy)

Answer: $|4.92 - 5| = 0.08$ pound; $0.08 \div 5 = 0.016 = 1.6\%$ off the label

6. Explain why accuracy is written with absolute value bars rather than plain subtraction. (Explain)

Answer: Accuracy is a DISTANCE from the true value, so it must never be negative – the bars guarantee a positive result no matter which value is larger.

7. A principal wants to name the “most improved” class. Class A rose from 62 to 74; Class B rose from 80 to 88. (Choose a Metric)

a. Find each class's raw point gain. Who improved more?

Answer: A: $74 - 62 = 12$; B: $88 - 80 = 8$. Class A improved more.

b. Find each class's percent gain. Who improved more?

Answer: A: $12/62 \approx 19.4\%$; B: $8/80 = 10\%$. Class A again.

c. Both metrics agree here. Why does that make the conclusion stronger?

Answer: When two independent metrics point the same direction, the result does not depend on which metric you happened to pick.

8. Error Analysis – Gus says a thermometer reading of 101.2°F when the true temperature is 101.0°F has an accuracy of -0.2°F . Identify his mistake. (Error Analysis)

Answer: He subtracted without absolute value and reported a negative distance. Accuracy = $|101.2 - 101.0| = 0.2^{\circ}\text{F}$.

9. A team scored 1848 points in 22 games. Find the points per game, then explain why a coach might prefer this metric over total points. (Metrics)

Answer: $1848 \div 22 = 84$ points per game. It allows fair comparison between teams that have played different numbers of games.

10. Describe a real situation where the mean would be a POOR model, and name a better metric. (Reason)

Answer: Sample: home prices on a street with one mansion – the outlier drags the mean far above a typical home. The median is the better metric.